



Conceptual modeling: Foundations, a historical perspective, and a vision for the future

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ABSTRACT

We recount the foundations of Conceptual Modeling in Computer Science, Philosophy and Cognitive Science and their implications on what are concepts, conceptualizations, and conceptual models. We then review the history of the field, considering earlier work by the three co-authors, and highlight some of the contributions that made it what it is. Finally, we propose three research directions whose solutions could advance the field and will hopefully be addressed in the future. Our study is intended to help to circumscribe and characterize the field. It draws ideas from Philosophy, Cognitive Science, Engineering and the Social Sciences, as well as several areas within Computer Science, including Programming languages, Artificial Intelligence, Databases, Software Engineering, and Information Systems Engineering.

1. Introduction

Every research field in Science and Engineering worth its salt is founded on some general principles. Such principles answer questions such as “What are the basic concepts in our discipline?”, “How are these concepts different from the ones used in other fields?”, “What are the research methods we use and do not use, and why?” and “What are open research problems to address?” Answers to such questions inform a field by pointing out relevant work in other disciplines from which to draw ideas and research results. They also suggest research questions to be explored and methods to be used in such explorations. In short, such principles provide the foundations that guide research. And these foundations need to be explicit and shared, worked out through meaningful discussion among members of the research community, conducted at research forums. Such a discussion can hopefully lead to more coherence in the research goals pursued by a research community, the research methods adopted, and the technical discussions held.

In a recent paper, Delcambre and colleagues [1] have started such a discussion by asking questions such as “What are conceptual models?”, “How are these different from models used in Computer Science and other disciplines?”, “Among models used in Computer Science, which are conceptual, and which are not?”. We applaud this initiative. Indeed, this is a long overdue discussion, necessary if Conceptual Modeling is to come to enjoy top academic status among related research fields, such as Knowledge Representation, Requirements Engineering, Formal Ontology, and Business Process Management.

In this paper, we first address the foundations of Conceptual Modeling by answering the question “What is a conceptual model?” while revisiting and extending [2], which studied the philosophical foundations of Conceptual Modeling and focused on what a conceptual model is and is not. We then circumscribe and characterize the field by reviewing its origins and history and discuss three seminal ideas that shaped the field. We close by presenting open research problems that may be worth studying in the future.

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Writing a complete history of the field is beyond the scope of what we aim to achieve here and would require a dedicated paper of its own. Our brief historical reconstruction is focused on one perspective albeit a broad one, which overlaps with multiple research fields within Computer Science, including Programming Languages, Artificial Intelligence (AI), Databases, Software Engineering (SE) and Information Systems Engineering (ISE).

In a recent paper [3] discussing (among other things) the intellectual structure of the field and what it reveals in terms of research traditions, methodological variations, or important themes, the authors identify four main clusters: (1) Conceptual modeling and databases; (2) Grammars and guidelines for conceptual modeling; (3) Requirements engineering and IS design methodologies; (4) Ontology constructs for conceptual modeling. We recognize ourselves as strongly connected to clusters (1), (3) and (4), which is corroborated by the authors of that paper. Accordingly, our historical and prospective perspectives here focus on these dimensions. We shall not address cluster (2), which reflects a vast and important tradition that goes back to the seminal work of [4,5] and influenced an entire school of thought that includes [6–18], among many others.

The remainder of the paper is organized as follows: Section 2 briefly discusses the foundations of conceptual modeling, including the very notion of concept as well as the related notions of conceptualization, abstraction and (conceptual) model (including ontologies). It also discusses what makes certain concepts (and, hence, also certain conceptualizations and models based on them) more stable and useful; Section 3 presents our historical reconstruction of the field with a focus on the aforementioned clusters; Section 4 presents our contribution for a future research agenda. In particular, we discuss three pathways: (1) investigating the explanatory role of conceptual models, (2) properly dealing with conceptual models as concrete (language) artifacts designed with a purpose and with specific requirements and quality attributes, (3) properly recognizing the social nature of conceptual models as artifacts, which are then co-constructed in a participatory manner involving other stakeholders beyond the modelers themselves; Section 5 presents some related work, i.e., work of other colleagues that pursue a project similar to the one we are set out to pursue here; Section 6 presents some final considerations.

2. Conceptualizations, conceptual models, and ontologies

2.1. What are concepts?

Conceptual modeling is about concepts, of course. But what are they? As we may expect, this question has been highly debated in Philosophy [19] as well as in Psychology (in particular, in Cognitive Psychology [20–26]). We present an abridged account that works for our purposes, according to which what underlies concepts, conceptualizations, and conceptual models are Carnapian intensions [27].

As human beings and sentient agents, we are exposed to the so-called physical reality, whose phenomena are studied, for instance, in physics, chemistry, or astronomy. We interact with such reality through our body and our limited perceptual systems. Throughout our life (crucially, during our first years), reality presents itself to us in multiple presentations, which are temporally atomic experiences. From these and with help from our social environment (family, school) we gradually learn certain invariances, which are cognitively relevant objects and events, forming what we call our cognitive domain. For each presentation (which includes everything we perceive), there is a corresponding world, which is a particular arrangement of relevant objects and events in that presentation. We also learn the cognitively relevant properties and relations involving the elements of the cognitive domain. These are intensional entities in Rudolf Carnap's sense [28], since they are rules to individuate instances of a property or a relation in each possible world and are therefore functions from possible worlds into actual relations in the cognitive domain. Such intensional entities may be called conceptual properties and relations, or simply concepts. Note that so-defined concepts are mental phenomena, they live neither in (external) reality nor in computers.

In summary, concepts reflect cognitively relevant regularities in a domain. They typically uncover universals in the world as repeatable patterns of features present in a multitude of individuals. They can be monadic concepts (e.g., Person) but also relational concepts or simply relations, also called associations, such as being married to, being the president of, being part of. According to empiricists (such as David Hume and John Locke), associations come about when concepts systematically co-occur in the experiences of a cognitive agent. For example, the concepts of 'Student' and 'Person' co-occur every time you encounter a student, so a Student/Person association is meaningful, likely and strong. Most proposals for approaches to conceptual modeling adopt such an associationist stance, including semantic networks, object-oriented models, and description logics. Finally, we can have individual concepts (such as the concept of 'Mick Jagger'). An individual concept can be formalized as a function from possible worlds to an individual (particular). For example, 'Mick Jagger' can be seen as a function that unifies multiple different individuals (corresponding to possible states of Mick Jagger) across possible worlds [29]. This function reflects a principle of identity that constrains those possible states as states of the very same individual. This principle of identity is supplied by the unique kind that individual instantiates, in this case the kind Person.

2.2. Conceptualizations, conceptual models, and ontologies

The view described in the previous section is basically the one presented by Guarino and Giaretta [30] and further clarified in [28,31], slightly extended here to stress the cognitive mechanism of learning concepts from experiences of reality. According to this view, a conceptualization is a triple $\langle D, W, R \rangle$, where D is a cognitive domain, W the set of all possible arrangements of that domain, and R is a set of conceptual properties and relations on D .

While conceptualizations are in the heads of people, conceptual models are information artifacts built with the purpose of representing (relevant aspects of) conceptualizations, chiefly for the purpose of communication, understanding and problem solving [32]. Computational ontologies are similar to conceptual models, since they are still primarily motivated by the needs of human communication, but they may be further defined as specifications (not just representations) of a conceptualization, in the sense that they aim at capturing as much as possible the intended meaning of the language used to talk about a conceptualization. Both conceptual models and ontologies require to fix a domain of discourse (aka domain), which in general will be a subset of the cognitive domain, and a vocabulary to talk of the elements of the domain as well as their relevant conceptual properties and relations. In general, a domain will cover a particular area of interest, such as Medicine, Banking, Food, Molecular Biology, etc. While the language constructs used in conceptual models are usually graphical primitives, as in UML, ontologies sometimes adopt a logical language which is often a variant of first-order logic, suitably weakened (as in OWL) to account for the tradeoff between expressivity and computational tractability.

Ontology-driven conceptual models lie at the intersection between conceptual models and ontologies. In this case, the modeling language constructs explicitly commit to a system of ontological categories in the following sense: (a) the modeling primitives reflect the ontological distinctions within that system of categories; (b) the constraints on how these primitives can be combined reflecting domain-independent and domain-specific ontological laws [33].

Conceptual models in Computer Science are (almost always) computational representations of conceptualizations. As such, they are built around computational models of universals and particulars. Conceptualizations can be constituted by monadic or relational concepts (corresponding to universals) as well as individual concepts (corresponding to particulars). Although most conceptual models represent monadic and relation concepts, domain-specific models (e.g., a subway map of Amsterdam) are constituted by individual concepts (e.g., Rokin station; the connection between the 50 and the 52 Lines) that instantiate general monadic (e.g., station, line) and dyadic concepts (parthood).

2.3. Conceptualizations and abstractions

It is a fundamental tenet of Cognitive Science and Philosophy of Mind that cognitive processes generate, use and transform mental representations. As discussed earlier, such representations are intensional, in the sense that they trace concepts across worlds (situations, circumstances) and are not limited to world-bound extensional classes of things as sets. Mental representations may be conceptual, such as thoughts, beliefs and desires, or non-conceptual, such as sensations [34]. For the purpose of brevity, we shall refer to conceptual mental representations as mental models and treat them as a synonym to conceptualizations.

Mental models can be seen as the “lens” through which a cognitive agent conceives an abstraction of the world. As we can observe in Fig. 1, which is a variant of the semiotic triangle of Ogden and Richards [35], the relation between conceptual models and reality is always mediated by a mental model (i.e., a conceptualization), which is aligned to human values and goals. For example, suppose we want to navigate a given city by using the subway system. By using concepts such as subway line, subway station, line direction, ordering of stations, intersection between lines, we can produce a mental model of a particular subway system. In producing such a model, concepts function as cognitive filters, allowing us to strip away properties that are not germane to the problem-at-hand (i.e., navigating the subway system), such as the width, luminosity and temperature of the tunnels.

Despite being dependent on cognitive agents, mental models however are not completely subjective, i.e., agents cannot choose to conceptualize the world in any arbitrary way. Indeed, mental models are articulated with basic concepts representing domain-independent ontological categories (e.g., objects, processes, events), which are to a large extent given, i.e., they are part of the ontology underlying human cognition, which is there exactly to afford the formation of stable and usable domain-specific concepts [21]. In other words, although our cognitive ontology and, hence, our mental models depend on our existence as the types of creature we are (with a particular history of influences by evolutionary forces), the former and certain aspects of the latter are not there as results of particular epistemic choices (e.g., we cannot help but see objects and processes in the world; we cannot help but reason using a sortal-supplied system of identity [24]). So, mental models (conceptualizations) are neither objective nor subjective, they can be said to be intersubjective or, to use a notion put forth by the philosopher John Searle [36], they are ontologically relative (since they depend on cognitive beings) but they are epistemically absolute (since they are to a large extent independent of our epistemic choices).

Although a conceptualization is the result of an abstraction process, its conceptual models—being information artifacts built chiefly for the purpose of communication and understanding — should include all elements that explain the ontological assumptions regarding the phenomena being observed. For example, consider Newton’s Second Law of Motion defined by the equation $Force = Mass * Acceleration$. Force can be thought of, in the gravitational case, as a relational complex connecting physical objects [37]; mass can be thought of as a quality of a physical object; acceleration as a quality of a process that is the manifestation of certain dispositions [38]. Now, none of these domain concepts (i.e., the physical objects, their qualities, the processes in which these objects participate, the qualities of those processes) are represented in that equation. The equation instead merely describes relations between values in measurement structures (abstract entities) associated with some individuals instantiating these concepts. So, from a conceptual point of view, that representation is not a conceptual model.

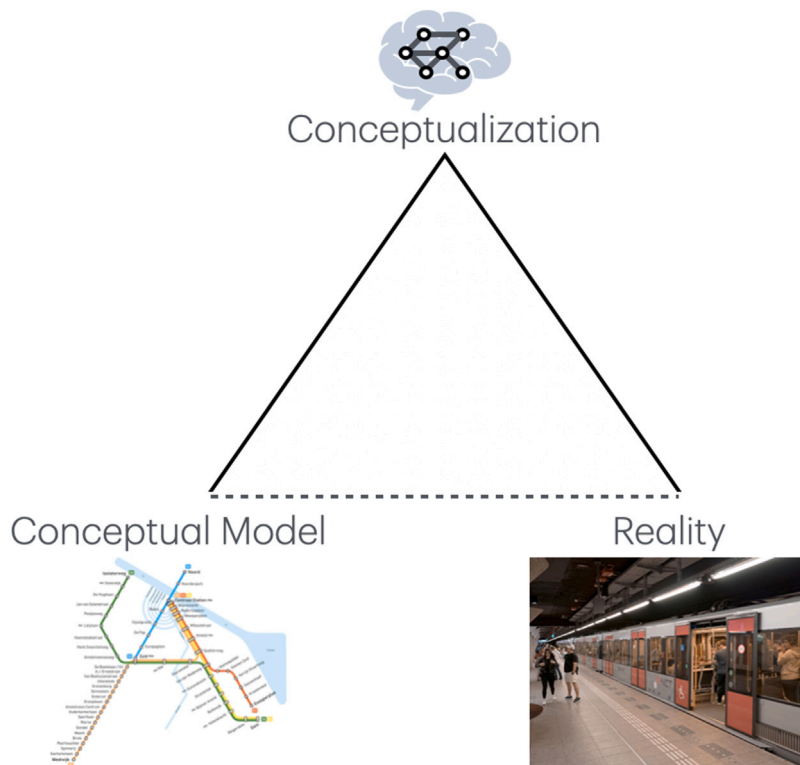


Fig. 1. The meaning triangle for conceptual models subway map by (by Michiel Straathof; subway photo by Fons Heijnsbroek).

2.4. What are ‘Useful’ concepts?

As described above, we subscribe to the view that concepts are mental representations of regularities (invariances) in the world. At this point, one might ask: are there any constraints on the way concepts are formed? What are ‘useful’ concepts? To put it another way: why is it that some concepts that describe and classify things in the domain are informative and useful, while others are vague, absurd or useless? This is a question studied by philosophers and psychologists (e.g., [39]), which does not concern the concepts belonging to a conceptualization, which are ‘useful’ by definition, but is especially relevant for the concepts explicitly introduced in a conceptual model, which in principle may be defined as arbitrary logical combinations of basic ones. The ‘Usefulness’ of such concepts has been defined in terms of two orthogonal properties: Naturalness and Coherence.

*Natural*¹ concepts [20] are formed out of genuine ontological categories, for example, ‘Person’, ‘Artifact’, ‘Student’, and ‘Planet’, but also ‘Object’, ‘Event’, ‘Relationship’. By contrast, unnatural concepts include those that cross ontological boundaries, such as the hypothetical concept of ‘thought or fish’, which has as instances thoughts or fishes, or the concept of ‘not a Dog’, which has as instances thoughts, fishes, and very many other dissimilar things. In his ‘Analytical Language of John Wilkins’, J. L. Borges [40] tells us about a fictitious alternative classification of animals that included categories such as ‘those that belong to the emperor’, ‘those that just broke a vase’, ‘the embalmed ones’, ‘the fabulous one’, etc. In discussing these examples, the psychologist Paul Bloom writes [41]: “Those that resemble flies from a distance’ is a logically possible way to group objects, but it is not how we naturally make sense of the world. No real language would have a noun for such a category. Real nouns capture something deep; *they refer to kinds of things* that are thought to share deep properties...” (our emphasis).

At this point, one could be tempted to make the alleged counterexample of the Dyirbal language of North Queensland, in Australia, which contains the term ‘balan’ that is used to talk, among other things, of “women, fire, and dangerous things”. The example became notorious in the homonymous book by the cognitive linguist George Lakoff [42]. This is, however, not a counterexample. The term ‘balan’ in Dyirbal (as well as the terms ‘bayi’, ‘balam’ and ‘bala’) is not a noun, but behaves as an article, marking a grammatical gender. Everything is therefore classified according to its gender, following some conceptual principles and rules that are often orthogonal to ontological classifications. For example, ‘bayi refers to masculinity or simply animateness, ‘balan’

¹ By natural concept here, we do not mean a concept that exists independently of human cognition and conceptualization. For example, there are multiple concepts of ‘Person’ (e.g., as human being, as cognitively capable human being, as legal person, etc.) that serve different social, legal and cognitive purposes. What we mean by natural is related to the notion of causal nexus connecting the properties characterizing a concept. This notion shall become clear throughout the paper.

to femininity; an important property is ‘harmfulness’ and, hence, although most fish is ‘baya’, dangerous fishes are classified as ‘balan’. To make an analogy, the term ‘das’ in German also marks a gender but we would not say that Germans have a single concept covering all entities described by a neuter name. On the contrary, certainly Germans and Dyirbal speakers do have concepts for the various nouns belonging to a specific gender, providing the individuation and identity conditions needed to deal with their instances in a competent way.

The reason why real languages do not have nouns for these concepts is the connection between count nouns and their ontological counterparts called sortals, in particular, substance sortals or kinds. An important issue is that kinds are those types that provide principles of individuation, counting and identity (both synchronically and diachronically) for the individuals it collects. Without the support of a sortal (ultimately, without the support of a kind) we cannot make any determinate statement of identity or counting (counting requires individuation and identity) [21,22,24,30,43]. Types that collect individuals of multiple kinds are called non-sortals. Notice that the gender ‘balan’ (like ‘das’) marks a noun class and, hence, technically it is not even a type that classifies individuals but a classifier for nouns. However, let us entertain for a moment that we have a category *balan** that classifies all instances of ‘Woman, Fire and Dangerous Things’. Not even in this case the categories *balan** and ‘those that resemble flies from a distance’ would be on equal footing: the former would be a non-sortal that presupposes that we have prior ways of individuating and identifying the individuals it collects (supported by the kinds fire, woman, and specializations of dangerous things); ‘those that resemble flies from a distance’ in contrast can perhaps merely count with a principle of individuation inherited from the kind ‘Fly’ but not with a principle of identity. In other words, one would not be able to know whether two instances of ‘those that resemble flies from a distance’ are the same or not and would not be able to know what changes an instance of this type can undergo and still be the same. These issues are not there for the instances of ‘Fire’ or ‘Woman’ that happen to be classified as *balan**. *Balan** is equivalent to our categories of ‘insurable item’ or ‘cultural heritage item’, except that the Dyirbal language would have different goals informing the formation of their non-sortals.

Besides being natural in the above sense, concepts should also be coherent [39]. They are coherent when they offer explanatory value for the domain, by presupposing a network of other concepts. For example, should the concept of winning team be just based on whether its instances are teams that have more wins than losses it would be basically tautologous, with a little, if any, explanatory value for the domain of sports. On the other hand, defining a winning team as a team whose instances, e.g., have: (a) several-star players, (b) in key positions, and (c) deploy state-of-the-art playing strategies, does have explanatory value for the domain: a team is a winning one because of (a)–(c) and it is therefore coherent.

Usefulness of concepts to their explanatory power has been linked to accepted scientific theories, but also to naïve theories [39], even wrong ones, as a basis for useful concept definitions. For example, a definition of the concept Storm that embraces Greek mythology where storms were caused by gods (e.g., Zeus, Apollo) and therefore have an attribute ‘godResponsible’ and sub-concepts ZeusStorm, ApolloStorm, etc., is coherent, though based on a false meteorological theory. The important aspect is that coherence of complex concepts defined in terms of multiple other concepts is derived from a causal nexus connecting those concepts. In other words, coherent concepts are not mere combinations of other concepts. Take for example the concept of a chair. In a merely descriptive approach, one could talk about the material constitution of the chair, the number of legs, the area of the seating surface, etc. However, all chairs have a purpose: to sit human beings. And their properties, such as having four legs, being constituted by certain material, and having a particular seating surface area are causally determined by their purpose.

These requirements for naturalness and explanatory power are strongly connected and make sense given that concepts are intended to enable useful conceptualizations of a domain. The concepts that are most natural and coherent are the concepts that provide essential properties for their instances as well as principles of identity and individuation for these individuals, i.e., the so-called sortal concepts and, in particular, *kinds* [43–46]. This is because, from a learning point of view, the most useful concepts are those that afford the most reliable transfer of inductive knowledge [20,43]. In other words, a useful concept is one that allows one to transfer to a newly encountered instance of that concept, all the properties one previously knows about other instances of that same concept. A (natural and nominal) kind is defined in terms of a maximally coherent nexus of causally connected properties that apply to individuals in all situations they exist. Moreover, it provides criteria for individuating and identifying the individuals to which these properties apply. For example, if somebody tells you about a ‘White or Orange’ thing (an unnatural and incoherent concept), what do you know about this thing? Almost nothing. If I tell you that it is a white thing (a more natural concept but with low explanatory value), what do you know now? Still only that it is white (and that it reflects light in a certain way). Now, if I tell you that it is an iPhone16, you will know how to individuate it, identify it, and by doing that, you would be able to transfer to this new object everything you know about iPhones in general and what you learned about this particular iPhone in previous encounters with it.²

As this discussion shows, our acquisition of domain-specific concepts does not occur in a vacuum, instead it is guided by some formal, domain-independent notions. Ontological notions such as identity, sortality, and essentiality [49] play indeed a fundamental

² In [47], the authors discuss the notion of the cognitive economy as a property of a concept in its ability to maximize information about its instances with the least cognitive effort [48]. According to them, to maximize this property, a classification scheme should focus on ‘meaningful’ classes, which they characterize as having meaning different from other existing classes. Although the authors do not define ‘meaningfulness’ in this passage, their reference to Rosch makes us think of her notion of Basic-Level types, which, as discussed in [43,49] can be thought as the cognitive counterpart of the ontological notion of sortal, in particular, of a kind. In [47], the authors also discuss the notion of Inference as a property of a concept concerning its ability to draw conclusions about its instances. Again, referring to Rosch, they claim Inference is possible because “the material objects of the world are perceived to possess high correlational structure”. Kinds satisfy this requirement of maximal discreteness (by definition, since all kinds are mutually disjoint [43,49]) and, as we discuss here, being maximally coherent nexuses of causally connected properties that apply to their instances essentially, they maximize the transfer of inductive knowledge about these instances given their classification under that kind.

role in guiding the acquisition of domain-specific concepts. Authors such as John Macnamara [21,22] have discussed at length the role of these ontological notions for cognition and, in particular, for supporting the learning of useful concepts (including individual concepts associated with proper names). Moreover, authors such as [25] provide evidence for the role of sortal concepts and, in particular, kinds for the cognitive understanding of individuation and identity.

In conclusion, a careful distinction must be made between concepts that belong to a conceptualization and the logical predicates and expressions that may represent them. Composite predicates resulting from disjunctions and/or negations of primitive predicates are incoherent concepts, as their instances have no common properties. ‘Useful’ concepts should be natural and have explanatory value.

2.5. Why concepts cannot be reduced to logical predicates?

Armstrong [50] and Bunge [51] convincingly argue that although concepts (universals) may be represented as logical predicates, it is not the case that any logical predicate represents a universal concept in a conceptualization. Consider, for example, a predicate $P(x) := C(x) \vee M(x)$, defined to be true for all objects x that have either an electrical charge or a mass. Let a and b be two individuals such that $P(a)$ and $P(b)$ hold. Now, knowing that P applies to both a and b , one cannot conclude that a and b have anything in common. Consequently, P is not a genuine universal. A further argument, according to Armstrong, is based on the link between universals and causality. If a particular instantiates a certain universal, then it may causally interact with the world in a certain way. Different universals determine different causal interaction behaviors. For instance, by virtue of having a certain mass, b can act upon a scale. Likewise, by virtue of having a certain charge, a can repel certain things. Now, were P a genuine property of a or b , it would add something to a 's and b 's behavior. But this is clearly not the case.

A similar case can be made for negative predicates. If C is a predicate denoting a universal, then, although the predicate $Q(x) := \neg C(x)$ would be perfectly acceptable in logics, it does not correspond to a genuine universal, since there is nothing necessarily in common between two things to which Q applies. Moreover, there is no common behavior among all the instances of Q . From a cognitive point of view, according to [52], only attributes representing substantial (i.e., genuine) properties will form convex regions (called quality regions) in a Conceptual Space. So, for example, while *Red* is a quality region in the color tridimensional space, *Red or Blue* or *not Red* are not. The philosopher David Lewis [53] names such genuine properties *natural properties*, as opposed to merely logico-linguistic predicates.

Finally, denying the one-to-one mapping between logical predicates and genuine properties does not deny that some negative or disjunctive properties can be co-extensional with genuine ones. For example, in a (simplistic) conceptualization of biological sex, one may assume that the concept *Person* is co-extensional with the disjunctive union of *Man* and *Woman*. However, there are indeed genuine properties captured by *Person*, properties that are common to all persons, such as common anatomic parts and biological processes. Analogously, *Man* is co-extensional with *Person* and *not Woman*. Again, *Man* is a genuine concept positing genuine properties to its instances. As the philosopher Kit Fine puts it [54], it is not in the very nature of (biological) men that they are not (biological) women. It just happens to be the case.

In conclusion, a careful distinction must be made between concepts that belong to a conceptualization and the logical predicates and expressions that may represent them. Composite predicates resulting from disjunctions and/or negations of primitive predicates are incoherent concepts, as their instances have no common properties.

In summary, if we put together this discussion with the one in the previous subsection we converge to the following dictum: ‘useful’ concepts should be natural and have explanatory value. Here, we focused on discussing these issues in terms of the genuine (causally powerful) properties bestowed on the instances of such concepts. In the previous subsection, we have focused on the usefulness of these concepts in providing explanatory value and affording inductive knowledge.

3. A historical perspective on conceptual modeling

The origins of Conceptual Modeling in Computer Science go back to the 60s. Ole-Johan Dahl and Kristen Nygaard proposed SIMULA 67, as a programming language for simulation programs [55]. The language was defined as an extension of Algol 60. The main extension consisted of the notion of a class that had instances, each with associated code so such instances were active, instead of passive data structures. The idea behind SIMULA 67 was that when you want to simulate a domain, for example barber shops, you define classes for the kinds of concepts whose behavior you will be simulating, such as barber shop, barber, customer and haircut. Accordingly, a simulation language should support the construction of models for a domain to be simulated that are both “direct” and “natural” for modelers: “direct” in the way they refer to domain elements and “natural” in that they correspond to our own conceptualization of that domain. For their proposal, Dahl and Nygaard have been acknowledged as fathers of Object-Oriented in Software Engineering and were honored with a Turing Award in 2001. SIMULA was followed by Smalltalk, developed at Xerox PARC, which formed a foundation for object-oriented programming and object-oriented modeling starting in the early 80s. The Unified Modeling Language (UML) constitutes a major achievement of this line of research, as it combined several modeling language proposals into one language for modeling software designs. UML has been extended into SysML (in its first version) to support modeling systems of systems, as opposed to mere software.

In AI, Ross Quillian [23] used in his PhD thesis semantic networks to model human memory. Semantic networks are directed labeled graphs where nodes denote concepts (more precisely, word senses), while labeled edges denote associations. His proposal was followed by many others in the early '70. Some semantic networks came with an interpreter that could draw inferences from the labels of links. Others had attached assertions or procedures to every node, capturing the semantics of the concept being

represented [56,57]. Among semantic network-based proposals for modeling languages, the KL-ONE knowledge representation language, proposed in the PhD thesis of Ron Brachman, stands out for its treatment of concepts, and the reasoning support provided. In that language, concepts are modeled as descriptions, while binary relationships are called roles. KL-ONE led to a family of languages known as Description Logics (DLs). The family includes the DL-based W3C standard known as the Web Ontology Language (OWL). OWL is considered a key technology in the Semantic Web landscape and vision.

In Databases, there were proposals for semantic data models, starting with Jean-Raymond Abrial proposing a Semantic Model [58], shortly followed by the Entity-Relationship Model of Peter Chen [59]. Both proposals — and many others that followed, e.g., [60,61] — were motivated by the desire to capture the contents of a database in a more, again, “direct and natural” way than the Relational Model that was the state-of-the-art in data modeling at the time. These proposals did not make claims that their modeling constructs represented concepts. However, all of them claimed that their models were “direct and natural” representations of the domain, or that they were able to “capture the true meaning of their domain” in the sense that they reflected our own conceptualization of a domain. The proposal of the Entity-Relationship Model and its immediate impact led to the launch of the Entity-Relationship Conference in 1979, later renamed Conceptual Modeling Conference, that celebrates this year its 44th anniversary.

In Software Engineering, Douglas Ross proposed in the mid-’70s the Structured Analysis and Design Technique (SADT) as a “language for communicating ideas” [62]. The technique was used to specify requirements for software systems. According to SADT, the domain consists of activities and data. Each activity consumes and produces data but also has data that control its execution but are neither consumed nor produced. Ross’ contributions include a modeling language that can capture both static and dynamic aspects of a domain. Ross insisted that SADT models should be intuitive and easy to understand, i.e., they correspond to the way we think of a domain. His proposal launched Requirements Engineering as a sub-field of Software Engineering and requirements models evolved from function-oriented to goal-oriented ones, e.g., KAOS [32,63].

Despite the recognition of the primacy of concepts over data structures in all these proposals, there was initially poor theoretical support for helping modelers choose the right set of concepts for a given domain, for making explicit the (real-world) semantics of these concepts, and for relating their modeling choices with other modeling choices that could be made about similar domains. Since the end of the 80s, researchers started systematically exploring the area of Ontology in Philosophy looking for such principles. These include Yair Wand and Ron Weber, as well as two of the authors of this paper. (a) Wand and Weber proposed in the early 90s a pioneering application of the philosophical system proposed by the philosopher Mario Bunge as a guiding framework for evaluating conceptual modeling languages [4,5]. (b) During the same period, the third author of this paper proposed the use of cutting-edge research in Philosophy, Linguistics and Cognitive Science as a way to define foundational theories and methodologies (e.g., the OntoClean methodology as a pioneering methodology for the evaluation of the ontological adequacy of taxonomic structures) [64]; (c) Leveraging and extending (b), the second author of this paper and his collaborators (including the third author) have proposed the first conceptual modeling language that was fully grounded on these foundational theories from ontology, cognitive science and linguistics [43,65]. The motivation behind these proposals (especially a and b) was that, if conceptual models are representations of conceptualizations of reality, then conceptual modeling languages and conceptual models should be guided by same notions that we use (frequently without awareness) to articulate our models of reality, i.e., by the notions constituting our cognitive ontology.

In summary, conceptual models were proposed and studied in at least four traditionally non-overlapping fields of Computer Science: Programming Languages, AI, Databases and Requirements Engineering. In all cases, they were proposed as models grounded in an intuitive, easy to understand, meaningful, direct and natural conceptualization of a domain.

It is worth recounting some of the seminal ideas that have served as foundations for and shaped the field. The first is the very notion of class proposed by Dahl and Nygaard as a programming language construct for modeling concepts about a domain. Without this construct, the modeler needed to resort to data structures such as records and lists to represent a concept, its behaviors and instances, rendering such representations anything but direct and natural. The second was the idea of Peter Chen, operationalized in the Entity-Relationship Model, that data are about a domain and when you are designing a database, you start by modeling that domain through the entities and relationships whose data will populate the database-to-be. The third was the idea proposed by the aforementioned researchers that conceptual modeling could systematically benefit from ontological foundations developed in philosophy and related areas.

4. A vision for the future

We present now three research ideas that follow from the definitions and implications in previous sections.

4.1. Conceptual models with explanatory value

Traditional conceptual modeling languages basically offer no means for explaining a domain. According to [66], models produced in these languages tend to be merely descriptive. Following the philosopher Bas van Fraassen, “explanations are over and above description” [67] as they need to, at the bare minimum, answer ‘know-why’ in addition to ‘know-that’ questions [68].

Explanations are receiving a significant amount of attention in AI and Software Engineering for the same reason: the systems that we build need to be trustworthy to their stakeholders, and an explanation capability is an important component of this trustworthiness requirement [69]. One of the reasons for this is that explanation is strongly connected to contestability: we need to understand a decision-making process (i.e., why a certain decision was taken) to be able to contest these decisions in case we

believe them to be inadequate. We expect the demand on conceptual modeling approaches for providing an adequate support for explainability to increase in the future.

In Cognitive Psychology, there is ample evidence (e.g., [20]) that the most reliable concepts are those whose characterizing properties are causally connected. In traditional conceptual modeling languages, however, concepts are defined as mere descriptive lists of features (see the discussion in Section 2 on conceptualizing chairs as merely descriptive lists of features versus disclosing the causal nexus that connects their characterizing properties). A modeling approach that can support explaining these causal connections and designing (or discovering) classes around these explanations would be a clear contribution to the field.

Every conceptual model makes an ontological commitment. However, in merely descriptive models, these commitments remain tacit in the mind of the modeler. According to [66], to explain a concept is to make explicit the ontological commitments behind its description. This is fundamental for guaranteeing the semantic interoperability of models as well as of the software and data based on them. [66] propose the first conceptual modeling method for systematically doing exactly that. This method, termed *ontological unpacking*, aims at revealing the truthmakers behind the propositions (truth-bearers) represented in a descriptive model. Truthmakers, in turn, represent what in the world is assumed to be responsible for the truth of that description. For example, while a merely descriptive model might represent that “Patients can be treated by one-to-many Hospitals”, an explanatory model would make explicit (among other things): (i) the truthmaker in virtue of which that relation holds and what is the nature of that truthmaker (e.g., a treatment contract binding patient and hospital at a given moment in time, or an event representing a historical connection between patient and hospital); (ii) how those truthmakers are brought to existence, how they are terminated, and in which ways they can change; (iii) how many entities are bound by a given instance of those truthmakers (e.g., whether several patients and/or several hospitals participate in the same treatment, or each treatment connects only one patient and one hospital but they can both participate in many treatments). Approaches that aim at systematically supporting this process of explaining the real-world semantics of models are also a great contribution to the field.

Explanations are always targeted at explanation-seekers, who have specific backgrounds and goals in mind. It becomes therefore essential to be able to elicit and explicitly represent what are the explanation requirements for a concept, i.e., what kinds of features of a concept need to be explained, at which level of abstraction and granularity, to what audience, and how. In this line of research, [70] proposes an interesting conceptual model for causes intended to address explanation requirements for software system actions.

To fully develop a conceptual modeling approach that systematically produces models that are explanatory (as opposed to merely descriptive), we need to fully develop formal systems of ontological categories aimed at accounting for all the notions that comprise our cognitive ontology, i.e., the system we use to articulate our conceptualizations (mental models) in the first place. Over the past three decades, there a handful of proposals for developing these systems. One that is particularly targeted at providing foundations for conceptual modeling is the Unified Foundational Ontology [71] and its associated conceptual modeling language OntoUML [43]. This ontology comprises micro-theories dealing with all essential domain-independent concepts used in conceptual modeling such as types and taxonomic structures, parthood, dependence, intrinsic and relational properties, attributes and attribute value spaces.

As demonstrated in [72,73], OntoUML is a pattern language whose actual modeling primitives are ontology design patterns representing these underlying theories. As discussed in depth [66], these patterns support the so-called Unificatory as well as the Metaphorical approaches to explanation, in which domain-specific phenomena are explained by a theoretical reduction to already known pattern or by a “metaphorical re-description of the domain of the explanandum” [74]. For example, a marriage is explained by mapping it to a particular guise of the legal relation pattern, thus, redescribing it as a particular type of legal contract composed of particular types of legal positions (duties, rights, powers, subjections, etc.) [75,76]. This explanatory strategy is also in line with the notion of explanation via *Pattern Subsumption* in Cognitive Psychology [77]. Experiments in this field provide evidence that explaining (via this mechanism of theoretical reduction or pattern subsumption [77]) guides learners to interpret what they are learning in terms of unifying patterns of regularities, which promotes the discovery of broad generalizations and the acquisition a deep understanding about the explained concepts *despite not having received any additional input from the world* [78]. As these authors demonstrate there, there is a clear influence of conceptual explanation in learning and reasoning.

Despite some important results [65], more foundational work is needed for dealing with some critical but less understood ontological categories (e.g., higher-order types, events, processes, situations). After understanding the nature of these categories, we need to understand how they can be incorporated into language systems and how they can be leveraged to support conceptual model explanations.

4.2. Conceptual models as artifacts

Engineering is about designing and building artifacts. In his classic collection of essays on artifacts [79], Herbert Simon postulates the principle that every artifact must have a purpose consisting of goals (aka requirements) before it is designed. Conceptual models are surely artifacts; therefore, they should come with their requirements before there is any model building. What would these requirements look like? And how does one fulfill them?

For software, also an artifact, requirements include functional requirements, for example, ‘request a meeting’ for a meeting scheduling software such as Doodle; also, quality (aka non-functional) requirements, such as ‘meetings shall be scheduled within two days from request’ and ‘meetings may have up to 100 invited participants’. Now, conceptual models do not have functional requirements, but they do have content requirements defined by competency questions [80] (or requests for explanation [67]) that must be answerable by the conceptual model once it is built. For example, for an enterprise architecture for a university department that includes staff (professors, administrators, teaching assistants, etc.), courses, academic processes etc., competency questions may

include “What courses were taught in the past 5 years?”, “What academic processes are in use right now?”, but also “What sub-categories of courses are there, if any?”, and also “What is the nature of the ‘studies-at’ relation between a student and a program?” The first two competency questions are about content at the instance level of the conceptual model, much like lookup queries answerable by a database. The third competency question is about the class level of the conceptual model and is intended to ensure that the conceptual model includes sub-classes for courses, such as Systems, Programming Languages and AI. The last question is a true request for explanation eliciting an answer about the very nature of the relation between the student and the program in which she is enrolled. To answer this question, one would need to understand the nature of the truthmaker associated with that relation (for example, with respect to different types of enrollments, each composed of different legal positions – e.g., rights, duties, liabilities, powers-, subject to different cardinality constraints, having different lifecycles, etc.).

Quality requirements are basically constraints on the qualities that apply for an informational artifact. There are proposals for such sets of qualities that may be adopted from international standards for software requirements specifications such as [81], or other proposals specifically for conceptual models [82]. For example, [82] proposes qualities for conceptual models along four dimensions:

- Domain – How well is the domain documented?
- Model – How valid is the model with respect to the domain described?
- Pragmatics – How comprehensive is the model for its intended audience?
- Language – How well is the modeling language used?

Each dimension is associated with different qualities and each quality with different metrics. For example, the Model dimension is associated with soundness (do all elements of the model correspond to elements of the domain?), and completeness (are all elements of the domain represented in the model?), and other qualities as well. In turn, soundness may be associated with the metric *#FalsePositives* that measures how many elements of the model should not be there, while *#FalseNegatives* measures how many elements of the domain are not represented in the model. Quality requirements can then be defined as constraints on one of more metrics associated with a quality, such as ‘*#FalsePositives(self.classes) ≤ 10%*’ (in words, the number of false positives among classes of the model is less than 10% of the total) also ‘*#FalseNegatives(self.classes) ≤ 1%*’ (in words, the total number of false negatives among classes of the model is less than 1%).

The requirements for a conceptual model can then be specified in terms of a set of competency questions and a set of quality constraints.

It is interesting that, unlike software, quality requirements at the class level would impact the model construction process. The example above calls for a conceptual model that has less than 10% false positives and less than one percent false negatives. This means that the modeler needs to check with almost all stakeholders that no concepts are missing, while only checking that a subset of the classes in the model represents concepts in the conceptualization.

This is just one proposal, and a naïve one at that. This research problem can be addressed by other technical proposals that are validated with case studies. For example, Conceptual Modeling can be addressed from the perspective of Design Science [79] and as such it would abide by the principles of that Science.

Still from the point of view of seeing models as artifacts, [83] propose a typology of modeling goals eliciting several value functions associated with different types of conceptual models. Examples include (a) supporting interventions in the world (prescriptive models); (b) domain understanding, conceptual clarification, meaning negotiation via model construction; (c) problem-solving (including decision-making) via model manipulation; (d) communication (including explanation, documentation). (b-d) are all supported by descriptive models; furthermore, the authors consider the case of creative models, i.e., models that bring about something in reality by their sheer existence in a context (e.g., models in wills, patent diagrams). This typology of modeling goals has received strong empirical support in [84]. Finally, [85] looks at models as organizational assets and, as such, analyze models as value objects (i.e., artifacts whose capacities can be enacted to satisfy the goals of the organization) [86], threat objects (i.e., artifacts whose vulnerabilities can contribute to denting the goals of the organization at hand) (ibid.), as well as objects at risk (i.e., objects that risk denting their value-adding capacities).

Finally, conceptual models are primarily targeted at humans and should, therefore, support these human users in the aforementioned tasks. In that respect, they should be pragmatically efficient and cognitively tractable. This requires that these models have (a) a cognitively adequate concrete syntax; (b) a way to make their semantics and ontological commitments transparent to both modelers and model users; (c) proper complexity management mechanisms, especially when these models grow to large dimensions (e.g., thousands of classes and relations [87]); (d) support for model and modeling language evolution.

Regarding (a), historically not enough attention has been paid to aspects related to the concrete (often visual) syntax of modeling languages. In the past 20 years, there has been an increase in publications in this area including (see, e.g., [88,89] for comprehensive reviews). However, most of the work in this area has been on the evaluation of visual concrete syntax, and only a few on proposing a theoretical account for the design of concrete syntax for conceptual modeling languages. We believe more research is necessary in that respect, leveraging on results from Cognitive Science (in particular, diagrammatic reasoning [90] and visual cognition [26]). Still regarding concrete syntax, an aspect that has been neglected in academia but is considered of great importance by practitioners is model layout [91]. In some areas, layout is particularly explored to convey pragmatic information [92,93]. We need more research to understand the science behind pragmatically efficient model layouts, especially in combination with (b) and (c). Furthermore, in a way that is analogous to how ontological semantics can systematically be leveraged in the design of concrete visual syntaxes [33], we need to investigate how ontology-based explanation could be systematically explored with visual strategies for explanation [94].

Regarding (b), as empirical research shows [95], modelers are almost always unaware of the semantic consequences of their modeling choices. Frequently, their models accept as valid possible interpretations that are not intended (under-constrained models) or exclude intended interpretations from being valid (over-constrained models). In recent years, we have seen important advances in using techniques for visualizing the possible interpretations of conceptual models. For example, [96] employs visual model-finding techniques for conceptual model validations. The strategy there is to generate visual representations of the possible interpretations of a conceptual model, so that the modelers and users can be confronted with its ontological commitments.

By using these techniques, [95] shows how anti-patterns (i.e., error-prone modeling structures) can be detected across models in different domains. Once these anti-patterns are known, they can be detected a priori and rectified in subsequent models.

As a final note on this validation strategy, we should point out that it offers a sort of execution semantics to conceptual models. The visualization of space of possible instances of a conceptual model brings several important benefits not only to the validation of specific models but also to conceptual model education. Programmers learn how to program not by reading the specification of programming languages but by experimenting with code (often other people's code!) and then executing it to see the results. In languages such as OntoUML, which offers this capability, we start seeing people downloading OntoUML models from its repository of models [97] and running it, changing the model, and running it again. Anecdotal evidence shows that this strategy imbues people with a fine understanding of the semantics of the language's constructs. This strategy of allowing for the execution of conceptual models (or at least the visualization of their populations) has also been advocated in [98,99]. We believe that pragmatically efficient execution semantics for conceptual modeling languages will be an active topic in the future.

Regarding (c), these techniques for model validation and anti-pattern detection rely on the premise that the modeler can understand the visual representations of the possible interpretations of their models. However, in complex domains, models can get prohibitively large, comprising thousands of classes and relations. In these scenarios, understanding the models (let alone the visualizations of their possible interpretations) become a cognitively intractable problem. In recent years, several techniques have been developed for automated modularization [100], viewpoint extraction [101], and model summarization [102]. However, this is still an area with open challenging issues [102,103]. Moreover, we are yet to see progress on how to support the validation of large-conceptual models, for example, by slicing the models in cognitively tractable chunks that could be validated in isolation but at the same time guaranteeing the preservation of the semantic properties of the whole.

Regarding (d), most conceptual models have been built in the past five decades without explicitly and systematically relying on theoretical foundations such as the ones advocated in this paper. But if these models are to be used as reliable support for tasks such as semantic interoperability, we need to cater for their evaluation by grounding them on such theoretical foundations. In recent years, we have witnessed the use of a combination of data-driven approaches (often using machine learning) with symbolic (logical) techniques to evolve conceptual models in this respect. For example, [104] employs the aforementioned visual validation techniques (i.e., visual model finding) together with a particular (symbolic) machine learning approach (Inductive Logic Programs) to learn missing constraints in conceptual models. In a different line of research, [105] employs Graph Convolutional Networks (GCNs) together with Large Language Models (LLMs) to provide automated grounding of classical conceptual models in a foundational ontology. Of course, still a lot must be done in terms of exploring the relation between Concept Modeling and sub-symbolic approaches to AI, in particular, LLMs. Understanding how to leverage LLMs as a tool to support different phases of the conceptual modeling process is still an open issue.

Beyond the evolution of individual conceptual models, we start seeing the emergence of data-driven approaches for evolving conceptual modeling languages. This is made possible by the existence of FAIR (i.e., Findable, Accessible, Interoperable, Reusable) repositories of conceptual models such as [97]. For example, [106] employs pattern-mining techniques to discover conceptual modeling patterns that recur in different models, done by different modelers and across different domains. High-Frequency patterns could eventually become part of the grammar of the associated modeling language. Sales and Guizzardi have in a series of papers [95,107,108] elicited patterns from ontology-driven conceptual models. Some of these patterns are stronger than mere "model smells" but they are real anti-patterns, i.e., modeling structures that should be avoided. Again, the knowledge of these structures may help language designers to evolve the grammar of modeling languages to include mechanisms to proscribe them. Finally, [49,109] have employed the notion of *systematic subversions* to cater for the evolution of a conceptual modeling language. Systematic subversion is a phenomenon in which different users of a language working in different domains and in a completely independent way bend the grammatical rules of the language in the same direction, thus inadvertently signaling to the designers of the language how the language should evolve to satisfy their modeling expressivity requirements. This view of a (language) artifact that is co-designed (and co-evolved) via conversations between designers and users is an example of the semiotic engineering process identified by [110].

As a final note, we would like to comment on the notion of concepts discussed here and how it relates to representations implemented in these sub-symbolic approaches, in particular, Deep Neural Networks. Authors such as [111] claim that Deep Learning "allows computers to learn from experience and understand the world in terms of a hierarchy of concepts, with each concept defined through its relation to simpler concepts". However, the notion of concept referred to there is very different from what we (as well as almost the entirety of the literature on the topic in Philosophy and Cognitive Science) mean by concept. As put forth by [112]: "representations in NNs are not really 'signs' that correspond to anything interpretable — but are distributed, correlative and continuous numeric values ... a hidden unit cannot on its own represent any object that is metaphysically meaningful". An even stronger case is made by [113], who shows that the whole project of abstraction in purely data-driven approaches (which embodied the idea of a bottom-up empiricist abstraction) necessarily generates representations of concepts that lead to a vicious (infinite and circular) regress in which concepts are reduced to mere amalgams of unexplainable similarities.

4.3. Conceptual models as social artifacts

The foundations of Conceptual Modeling postulate that a conceptual model represents the mental models of a cognitive agent. However, that cognitive agent may be an individual, or a social agent, such as a team of designers or a corporation. For example, many corporations maintain an approved list of business processes and have procedures in place for updating that list. This list constitutes a shared model for all members of the social agent. In fact, for conceptual models intended to capture a design, it is almost always the case that the cognitive agent is a social one, consisting of stakeholders and a design team.

Unfortunately, in design projects the social agent does not have a shared model at the beginning of the design process, this only comes about when the design is completed. This poses a challenge: How do we build a shared conceptual model for a social agent given the individual views of the agent's members? Originally, Philosophy going back to Socrates and Plato, and more recently the Social Sciences with Hegel, Marx and others, have answered this question with dialectic theories [114]. These are theories that propose dialectic processes of thesis-antithesis-synthesis for arriving to a shared solution to a given design problem.

These considerations suggest a dialectic process for building a shared conceptual model for a social agent. For example, to build an enterprise architecture for an academic department, the modeler may first consider the view of the director by asking questions such as "What programs are offered by the department?", then check the model that results from this discussion with other administrators who either remove elements of the model and/or add new ones and continue with other stakeholders until there is convergence and no stakeholder has any further changes to make to the conceptual model that has resulted from the process so far. Following this research direction can result in novel methods for building conceptual models. In [115], the authors propose such a method for building requirements models.

Other approaches along these lines look at the design of artifacts from communication [110] and negotiation [116] points of view. From the point of view of communication, [110] see artifacts as a dialog between designer and user: the modeler represents in the model choices that they deemed to be adequate to satisfy the goals of the user. The user, on the other hand, might bend the rules of model (and modeling language) to communicate with the modeler a deviation of that relation, i.e., to communicate misconceptions about their goals or about the capacity of those model elements to satisfy those goals. In contrast, from the point of view of negotiation, [116], for example, proposes an approach in which a collaborative team of modelers exchange modeling acts in order to co-design models.

5. Related work

[1] presents a reference framework for researchers and practitioners to position their work in the field. The positions we defend in this paper are very much aligned with many of the aspects represented therein. By capturing the opinion of a large number of experienced researchers in the field, the framework also highlights that a "conceptual model is an artifact", thus, built with a purpose, that conceptual models should "expresses the semantics of the scenario of interest", and that the purposes of a conceptual model include to: "answer questions about a domain, improve understanding, and promote knowledge sharing; expose (the author of a conceptual model's) assumptions about a domain; promote communication among people developing a conceptual model, or among people who (later) use a conceptual model". Finally, the authors also acknowledge that conceptual models can exist at different "genericity" levels, i.e., they "may be generic and thus intended for use in any application domain" or be domain-specific.

As discussed in [2], we agree with many of the points put forth by our colleagues, including the ones highlighted above. However, our approach here is very different in method from the contribution of our colleagues, albeit complementary to theirs. Their main objective was to systematize several characteristics attributed to conceptual models by consulting the literature as well as experts in the field. In contrast, we take here a stance in addressing the questions of why conceptual models have the characteristics they have. To do that, we built on several from the literature of several areas in computer science concerned with the constructions of conceptual models, as well as theories coming from the areas of Philosophy, Logics and Cognitive Science. Furthermore, our historical characterization of the field seems to be broader, e.g., discussing object-oriented models going back to Simula 67 or Description Logics going back to KL-ONE. Finally, unlike our colleagues we propose a prospective agenda for the field.

In [117], Mayr and Thalheim propose a triptych of Conceptual Modeling as a framework for better understanding of the field. The triptych is akin to our meaning triangle and Ogden and Richard's semiotic triangle, but it includes an extended linguistic dimension that goes beyond what we explored here. [117] also addresses the question of what is and is not a conceptual model and we agree with some of their answers, but not all.

Even prior to that paper, in [118], Thalheim analyzes and systematizes some of the notions related to conceptual models and discuss what he judged to be some of its essential ingredients. Like our previous paper [2], he also analyzes several propositions typically associated with conceptual models, some of which are true (and in line with our views) and some of which are false — he uses the term "Myths of Conceptual Modeling". Here, instead, we focus on providing a characterization of the notions of concepts, conceptualization and their representations (i.e., models). We do, nonetheless, explicitly dispense with the misconceptions that concepts are reducible to logical predicates, that all concepts are equally useful, and that all models can be put on the same footing (descriptive and explanatory models alike). Thalheim also elaborates on the nature of the notions of concepts and conceptualizations, albeit in a less formal manner. Finally, Thalheim also elaborates on several research questions. Differently from our proposals for research directions for the field, these refer to fundamental questions such as: "which functions have conceptual models in which utilization scenarios?", or "what is the role of the community of practice in conceptual modeling?". In any case, we highlight a few additional important and shared points touched by the paper but which we have elaborated here to higher level of details: conceptual models as artifacts (instruments) including the need for abstraction and refinement mechanisms in which the same model can be

presented to different stakeholders in different versions; the idea of a conceptual model as resulting from a negotiation process; the notion of a ‘deep model’ as a set of a prioristic assumptions (including of ontological guise) that guide the elaboration of a target model with a certain utilization scenarios in mind; the role of perception as a source for conceptual models.

[119] acknowledges the influence of Philosophy on Conceptual Modeling on topics such as Ontology, Epistemology and Logic, but also argues for principled pragmatism in adopting ideas from these fields, meaning that attention should be paid to the intended application of the ideas, rather than philosophical tenets.

[120] proposes six principles that should guide the design of conceptual modeling languages and tools: flexibility, accessibility, ubiquity, minimalism, primitivism, and modularity. These principles should serve as quality requirements that guide the design languages and tools and can move the field forward in achieving universal conceptual modeling. This paper also presents a list of research opportunities derived from the possible application of the principles.

Finally, Proper and several colleagues (including the second author of this paper), have approached the problem of characterizing conceptual modeling from different angles. For example, in [121], he and colleagues propose a conceptual modeling process defined as a dialog among stakeholders that shares many features with our proposal in section 4.3 – without making the link to the literature on dialectics. Moreover, in [122], together with Jan Schoonderbeek, they propose an ontology (in OntoUML) of the domain of modeling itself, with a particular focus on Enterprise Architecture Models. Their proposed ontology (which is also organized around Ogden and Richard’s triangle) elaborates on notions such as concept and conceptual domain models, and how the latter relate to issues such as (a) their cost-effectiveness, and (b) their cognitive purposes acknowledged by a collective agent. These characteristics (a) and (b) of models highlight their status as artifacts and, in particular, social-linguistic artifacts. The notion of domain models in this ontology is aligned with the one proposed in [83,123,124], which share many of the tenets we defend here (e.g., the perspectives discussed in sections 4.2 and 4.3). Their ontology, however, does not provide a formal characterization of these notions of concepts or conceptualization (beyond the one derived from the formal semantics of OntoUML).

6. Final considerations

We have attempted to characterize the foundations of the research field of Conceptual Modeling and circumscribe its boundaries and overlapping areas with other fields. The foundations have been adopted mostly from Cognitive Science and Philosophy, specifically Philosophy of Mind and Ontology. Our circumscription of the field is based on a short account of the history of the field, going back almost 60 years.

Conceptual Modeling is a core research topic for Computer Science, offering concepts, tools and techniques for building models for a host of other fields, including AI, Databases, Software Engineering, Requirements Engineering, Business Process Management, and Enterprise Architectures. As such, we believe that it should be taught at the undergraduate and graduate level as a subject in its own right, rather than through a few lectures on UML or the Entity Relationship Model at a software engineering or a database course.

Last, but not least, Conceptual Modeling is a field with a bright future as more and more areas of Computer Science and related disciplines discover the usefulness of models and and hopefully opt for their explicit, systematic and well-founded engineering. We have presented some open research problems that we consider important, but there are countless others that can and should be pursued. This means that this host journal has a bright future too, for it has served well the Conceptual Modeling community as the leading journal venue for our field.

CRedit authorship contribution statement

John Mylopoulos: Writing – review & editing, Conceptualization, Formal analysis, Writing – original draft. **Giancarlo Guizzardi:** Formal analysis, Writing – original draft, Writing – review & editing, Conceptualization. **Nicola Guarino:** Writing – original draft, Writing – review & editing, Conceptualization, Formal analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: None of the authors have received support reports was provided by No support received from the authors’ institutions or other organizations. None of the authors received financial support from any organization reports a relationship with None of the authors received financial support from any organization that includes:. There are no further relationships or activities to declare for any of the authors. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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